WORKED SOLUTION

STEP I 2009, Question 1

Pure • Counting Divisors

> **STEP I 2009, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/09-s1-q1

The key idea

Every factor of $p^a q^b$ (with p, q distinct primes) is $p^i q^j$ with $0 \leq i \leq a$ and $0 \leq j \leq b$, and different choices of (i, j) give different factors by unique factorisation. So the number of factors is a *product* $(a+1)(b+1)$, and the same holds with any number of primes. Counting proper factors just means subtracting 2, and both parts become questions about writing a number as a product.

Solution

Write $d(N)$ for the number of positive factors of N, including 1 and N. If

$$N = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

with distinct primes p_i and $a_i \geq 1$, then a positive factor of N is $p_1^{e_1} \cdots p_k^{e_k}$ with $0 \leq e_i \leq a_i$; there are $a_i + 1$ choices for each e_i, and by unique factorisation distinct choices give distinct factors. Hence

$$d(N) = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1).$$

For $N > 1$, the factors 1 and N are different, so the number of proper factors is $d(N) - 2$.

Part (i)

For $3^2 \times 5^3$ we get $d = (2+1)(3+1) = 12$, so there are $12 - 2 = 10$ proper factors, as required.

Now take $3^m \times 5^n$. For this to be an integer with a proper factor at all we need $m, n \geq 0$ (a negative power of 3 or 5 cannot be cancelled by the other prime). The factors are $3^i 5^j$ with $0 \leq i \leq m$, $0 \leq j \leq n$, so $d = (m+1)(n+1)$ (this also covers $m = 0$ or $n = 0$). We need

$$(m+1)(n+1) = 12,$$

that is, an ordered factorisation of 12 into two positive integers. These are

$$1 \times 12, \quad 2 \times 6, \quad 3 \times 4, \quad 4 \times 3, \quad 6 \times 2, \quad 12 \times 1,$$

giving

$$(m, n) = (0, 11),\ (1, 5),\ (2, 3),\ (3, 2),\ (5, 1),\ (11, 0).$$

Each gives a different integer, again by unique factorisation. One of them, $(2, 3)$, is the given number, so there are exactly **5 other** integers of this form with 10 proper factors: 5^{11}, 3×5^5, $3^3 \times 5^2$, $3^5 \times 5$ and 3^{11}.

Part (ii)

We need $d(N) = 426 + 2 = 428$, and

$$428 = 2^2 \times 107,$$

where 107 is prime (it is not divisible by 2, 3, 5 or 7, and $11^2 > 107$).

? **Ask yourself** Which shapes of prime factorisation can N possibly have?

The numbers $a_i + 1$ are each at least 2 and multiply to 428. Since 107 is prime, exactly one of them is a multiple of 107, and it is 107, 214 or 428; the remaining ones multiply to 4, 2 or 1 respectively. Ignoring order, the possibilities for the list of $a_i + 1$ are

$$\{428\}, \qquad \{214, 2\}, \qquad \{107, 4\}, \qquad \{107, 2, 2\},$$

so the multiset of exponents is

$$\{427\}, \qquad \{213, 1\}, \qquad \{106, 3\}, \qquad \{106, 1, 1\}.$$

? **Ask yourself** Given the exponents, which primes should carry them?

To make N as small as possible with a given list of exponents, we should use the smallest primes, and put the largest exponent on the smallest prime. Using the smallest available primes is clearly best: replacing a prime by a smaller unused one makes N smaller. And if $p < q$ but p carries exponent a and q carries exponent b with $a < b$, then swapping them multiplies N by

$$\frac{p^b q^a}{p^a q^b} = \left(\frac{p}{q}\right)^{b-a} < 1,$$

so the swap makes N smaller. So the best candidate of each shape is

$$2^{427}, \qquad 2^{213} \times 3, \qquad 2^{106} \times 3^3, \qquad 2^{106} \times 3 \times 5.$$

Comparing them:

- $2^{106} \times 3 \times 5 = 2^{106} \times 15 < 2^{106} \times 27 = 2^{106} \times 3^3$;

- $2^{106} \times 27 < 2^{106} \times 2^5 = 2^{111}$, which is far less than $2^{213} \times 3$ and 2^{427}.

Hence

$$N = 2^{106} \times 3 \times 5.$$

As a check, $d(N) = 107 \times 2 \times 2 = 428$, so N has 426 proper factors.

Checks

- For a small case, list the factors of $3 \times 5^5 = 9375$ directly: $3^i 5^j$ with $i \in \{0,1\}$, $j \in \{0,\ldots,5\}$ gives 12 factors, so 10 proper ones.
- A short computer search over all exponent patterns $2^a 3^b 5^c 7^d$ with $(a+1)(b+1)(c+1)(d+1) = 428$ confirms that the smallest is $2^{106} \times 15 \approx 1.22 \times 10^{33}$, with $2^{106} \times 27$ next.

Common slips

- Answering 6 in part (i): the question asks for the *other* integers, so the given one is excluded.
- Forgetting the cases $m = 0$ or $n = 0$ (that is, 5^{11} and 3^{11}), which are of the required form.
- Assuming the answer to part (ii) must use as many primes as possible: the primality of 107 forces one huge exponent, and only the split of the factor 4 is a real choice.

Beyond STEP

The function $d(N)$ is a *multiplicative* function: $d(mn) = d(m)d(n)$ whenever m and n are coprime, and that is exactly the product formula used here. Numbers that have more factors than every smaller number are Ramanujan's *highly composite numbers*, and the exponent-swapping argument in part (ii) is the first step in his description of them. Davenport's *The Higher Arithmetic* is a good place to read about unique factorisation and arithmetic functions.

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